

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Intelligent Dynamic Traffic Signal Control System

Abstract

Background & Problem Statement: Traffic congestion is a growing challenge in urban areas due to the increasing number of vehicles and the use of fixed-time traffic signal systems. Conventional traffic lights operate on predefined timers without considering real-time traffic density, resulting in unnecessary waiting time, fuel wastage, increased air pollution, and delays for emergency services. These issues are particularly severe near hospitals, airports, schools, railway stations, and commercial areas where traffic patterns vary significantly throughout the day..

Target Users / Stakeholders: The proposed system is designed for traffic management authorities, municipal corporations, smart city administrators, traffic police, emergency service providers, and daily commuters. These stakeholders require an efficient traffic control system that can dynamically respond to changing traffic conditions and improve overall road efficiency.

Existing Gap & Evidence: Current traffic management systems primarily rely on fixed signal timings or manual intervention, making them ineffective during peak hours or unexpected traffic congestion. Recent studies in Intelligent Transportation Systems (ITS) and reports from urban traffic management authorities indicate that adaptive signal control using Artificial Intelligence can significantly reduce vehicle waiting time, congestion, and fuel consumption. However, many existing solutions focus only on vehicle counting and fail to incorporate both location-based and time-based traffic priorities.

Related Work / Competitive Landscape: Existing AI-based traffic management systems use computer vision and deep learning models such as YOLO for vehicle detection and density estimation. While these systems provide adaptive signal control, most do not integrate traffic density analysis with priority management for critical locations such as hospitals and airports. The proposed system addresses this limitation by combining real-time vehicle detection, location-aware priority, and peak-hour scheduling into a unified traffic control framework.

Proposed Approach: The proposed project, INTELLIGENT DYNAMIC TRAFFIC SIGNAL CONTROL SYSTEM : An AI-Based Adaptive Traffic Signal Controller Using Real-Time Vehicle Detection and Intelligent Priority Management, uses cameras installed at the North, South, East, and West directions of a road intersection. An AI-based object detection model identifies and counts vehicles in each lane, estimates traffic density, and dynamically adjusts green signal duration. The system also considers predefined time schedules and priority zones to optimize traffic flow more effectively.

Scope & Boundaries: The project focuses on a single traffic intersection with camera-based vehicle detection, adaptive signal timing, and time- and location-based priority management. It does not include autonomous vehicle communication, multi-intersection synchronization, or direct deployment on physical traffic infrastructure.

MAJOR PROJECT REGISTRATION FORM

Date:

Expected Outcome / Success Criteria: The proposed system aims to reduce average vehicle waiting time, improve traffic flow, and minimize congestion compared to conventional fixed-time traffic signals. Success will be measured through traffic simulation by evaluating vehicle queue length, average waiting time, and signal utilization efficiency under different traffic conditions.

References (Category specific)

1. [1] H. Li, Y. Wang, Z. Chen, and X. Liu, "Deep Learning-Based Intelligent Traffic Signal Control: A Comprehensive Review," *IEEE Access*, vol. 12, pp. 45892–45918, 2024.
2. [2] S. Kumar, R. Sharma, and P. Singh, "AI-Driven Adaptive Traffic Signal Control Using Computer Vision," in *Proceedings of the IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2023, pp. 2145–2151.
3. [3] Y. Zhang, L. Zhao, and M. Chen, "Computer Vision-Based Traffic Density Estimation for Smart Cities," *Sensors*, vol. 23, no. 15, pp. 1–21, 2023.
4. [4] J. Redmon and A. Farhadi, "YOLO: Real-Time Object Detection," Available: <https://pjreddie.com/darknet/yolo/>. (Reference for the YOLO object detection framework used in vehicle detection.)
5. [5] Ministry of Road Transport and Highways (MoRTH), *Road Transport Year Book 2023–2024*, Government of India, New Delhi, India, 2024.
6. [6] Ultralytics, *YOLO11 Documentation*, 2024. Available: <https://docs.ultralytics.com/>



VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

Permanently Affiliated to JNTU Kakinada, Approved by AICTE
Accredited by NAAC with 'A' Grade, ISO 9001:2008 Certified
Nambur, Pedakakani (M), Guntur (Dt) - 522508

Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1.	23BQ1A61B9	RETTEDI TRIPURNA	
2.	24BQ5A6113	SHAIK CHINA KASIM	
3.	23BQ1A61C4	SANAPALA JASWANTH	
4.	23BQ1A61E6	VANGA RAGHURAMI REDDY	

Project Guide

Name: A.JANARDHANA RAO

Signature:

MAJOR PROJECT REGISTRATION FORM

Date:

INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.